

Polyplets with a Single Corner-Pinch

A corner-pinch contact is a pair of cells touching only at a corner: the two cells sit diagonally adjacent and both cells that would fill the gap between them along the grid axes are absent, so their only contact is the shared corner.

$a(n)$ is the number of free polyplets of n cells (king-move connected, counted up to rotation and reflection) with exactly one such contact. For $n = 2$ and 3 there is a single shape, shown centered. For $n = 4, 5$ EVERY distinct shape is drawn in a grid under one $a(n)$ heading, each labelled with its bounding box, so the whole family the count measures is visible – the shapes are genuinely different, not rotations of one another. Ordinary cells are red; the two cells of the corner-pinch pair are blue.

$$a(2) = 1, a(3) = 1, a(4) = 6, a(5) = 20$$

Computed by Peter Exley, Jun 2026

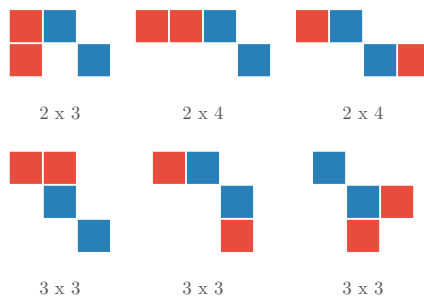
$a(2) = 1$ [PROVED] the only shape, 2 x 2 bbox



$a(3) = 1$ [PROVED] the only shape, 2 x 3 bbox



$a(4) = 6$ [PROVED] all 6 shapes, bbox under each



$a(5) = 20$ [PROVED] all 20 shapes, bbox under each

